

SHORE

2008

Trial HSC Examination

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided on the back page of this question paper
- All necessary working should be shown in every question

**DO NOT REMOVE THIS PAPER FROM
THE EXAMINATION ROOM**

Student Number:

Set: 12ME2-1 (FES)

Total marks – 120

- Attempt Questions 1 – 8
- All questions are of equal value
- Start each question in a new writing booklet
- Write your examination number on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

Blank Page

Total Marks – 120
Attempt Questions 1 – 8
All Questions are of equal value

Begin each question on a NEW BOOKLET, writing your name and question number at the top of the page. Extra booklets are available.

Question 1 (15 marks) Use a SEPARATE writing booklet

Marks

(a) Find $\int (e^x + e^{-x})^2 dx$ 2

(b) Find $\int \frac{2x^2 - 2x + 1}{(x - 2)(x^2 + 1)} dx$ 3

(c) Use the substitution $t = \tan \frac{x}{2}$ to evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{2 + \sin x} dx$. 5

(d) Let $I_n = \int_0^1 \frac{1}{(1 + x^2)^n} dx$; $n=1,2,3,\dots$

(i) Show that $I_{n+1} = \frac{2n-1}{2n} I_n + \frac{1}{n \cdot 2^{n+1}}$; $n=1,2,3,\dots$ 3

(ii) Hence evaluate $\int_0^1 \frac{1}{(1 + x^2)^3} dx$. 2

P.T.O.....

Question 2 Use a SEPARATE writing booklet

Marks

(a) $z_1 = 1 + i$ and $z_2 = \sqrt{3} - i$.

(i) Find $\frac{z_1}{z_2}$ in the form $a + ib$ where a and b are real.

1

(ii) Write z_1 and z_2 in modulus – argument form.

2

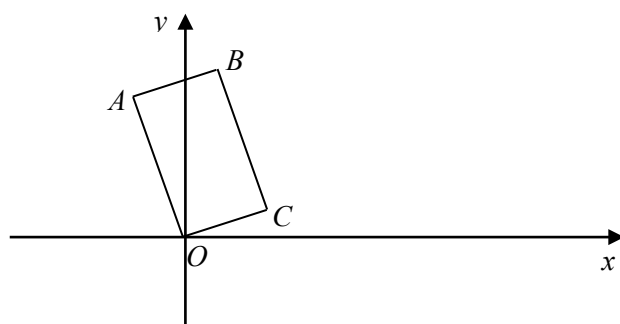
(iii) By equating equivalent expressions for $\frac{z_1}{z_2}$, write $\cos \frac{5\pi}{12}$ as a surd.

1

(iv) Explain why there is no positive integer n such $z_1 z_2^n$ is real.

2

(b)



The points $OABC$ are the vertices of a rectangle on the Argand diagram with $|OA| = 2|OC|$. If OC represents the complex number $p + iq$, write down the complex numbers represented by:

(i) $\vec{OA}$

1

(ii) $\vec{OB}$

1

(iii) $\vec{BC}$

1

(iv) $\vec{AC}$

1

(c) (i) If $z = \cos \theta + i \sin \theta$, explain why $z^n + z^{-n} = 2 \cos n\theta$ and $z^n - z^{-n} = 2i \sin n\theta$ for positive integers n .

2

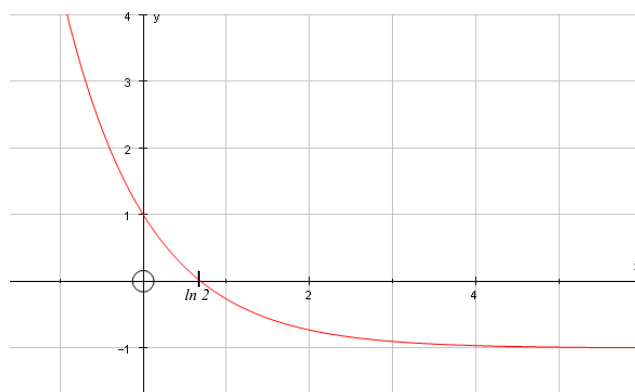
(ii) By considering the Binomial expansions of $(z + z^{-1})^3$ and $(z - z^{-1})^3$, show that $4(\cos^3 \theta + \sin^3 \theta) = (\cos 3\theta - \sin 3\theta) + 3(\cos \theta + \sin \theta)$.

3

Question 3 **Use a SEPARATE writing booklet**

Marks

(a)



The diagram shows the graph of $f(x) = 2e^{-x} - 1$. On separate diagrams sketch the following graphs, showing the intercepts on the axes and the equations of any asymptotes:

(i) $y = |f(x)|$. **1**

(ii) $y = \{f(x)\}^2$. **1**

(iii) $y = \frac{1}{f(x)}$. **2**

(iv) $y = \ln \{f(x)\}$. **1**

(b) Consider the curve $y^2 = x^4(4+x)$

(i) Sketch the curve. **2**

(ii) Find the area of the loop of the curve from $x = -4$ to $x = 0$. **3**

(c) The roots of $x^3 - 3x^2 - 2x + 4 = 0$ are α, β and γ . Answer the following without finding the actual values of α, β and γ .

(i) Find a cubic equation whose roots are α^2, β^2 and γ^2 . **2**

(ii) Hence or otherwise find the value of $\alpha^2 + \beta^2 + \gamma^2$. **1**

(iii) Determine the value of $\alpha^3 + \beta^3 + \gamma^3$. **2**

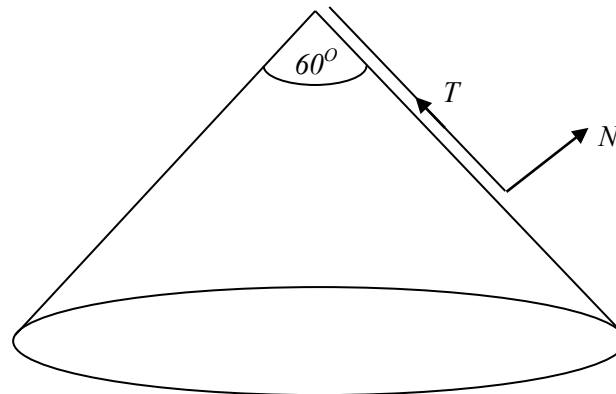
P.T.O.....

- (a) Given $|z + i| \leq 2$ and $0 \leq \arg(z + 1) \leq \frac{\pi}{4}$. Sketch the region in an Argand diagram which contains the point P representing z . 3
- (b) Consider the five 5th roots of unity.
- (i) Solve $z^5 - 1 = 0$ over the complex field giving your answers in modulus-argument form. 3
- (ii) Hence express $z^5 - 1$ as the product of real linear and quadratic factors. 3
- (iii) Write down the complex roots of $z^4 + z^3 + z^2 + z + 1 = 0$ giving your answers in modulus-argument form. 1
- (iv) Hence prove that $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$. 2
- (c) If $x > 0$ and $y > 0$ prove that $\frac{1}{x} + \frac{1}{y} \geq \frac{4}{x+y}$. 3

- (a) A mass of 3kg is attached to the vertex of a cone of vertical angle 60° by an elastic string of length 1 metre. The mass is moving in a horizontal circle on the curved, frictionless surface of the cone. Acceleration due to gravity is 10 m s^{-2} .

T = Tension in string

N = Normal reaction of the cone surface on the mass.



Not to Scale

- (i) If the mass is moving at a speed of 1 m s^{-1} , by resolving forces vertically and horizontally find the values of T and N. 4
- (ii) What is the maximum speed of the particle for it to just remain on the cone's surface and what will be the string's tension at this time? 4
- (b) A mass of 1 kg is allowed to fall under gravity from rest at the surface of a medium in which the retardation on the mass is proportional to the distance fallen (x). In other words, the net force for this motion is $g - kx$ Newtons with the downward direction as positive.
- (i) Show that it falls $\frac{2g}{k}$ metres before it becomes stationary. 2
- (ii) Show that the displacement equation in terms of t is given by: 5
- $$x = \frac{g}{k} \left(\sin \left(\frac{2\sqrt{k} t - \pi}{2} \right) + 1 \right)$$

P.T.O.....

(a) Consider the complex number z which satisfies $|z| = 1$.

(i) Using double angle trigonometric identities show that:

2

$$1 + \cos \alpha + i \sin \alpha \equiv 2 \cos \frac{1}{2} \alpha (\cos \frac{1}{2} \alpha + i \sin \frac{1}{2} \alpha).$$

(ii) If $z = \cos \theta + i \sin \theta$, $-\pi < \theta \leq \pi$, write $1 + z^2$ in terms

3

of $\cos \theta$ and $\sin \theta$. Hence deduce that if in an Argand diagram, points A and B represent z and $1 + z^2$ respectively, then A , B and O are collinear, where O is the origin. State the values of θ such that B lies on the interval OA .

(b) A solid shape has as its base the parabola $y = x^2$ in the XY plane.

5

Sections taken perpendicular to the axis of the parabola (i.e. perpendicular to the y -axis) are equilateral triangles. Using the method of slicing determine the volume of the solid, if the length of the axis of the parabola is 16cm.

(c) With θ increasing, the point P given by $(r \cos \theta, r \sin \theta)$ is moving in a circular motion

5

about O at a distance of r units from O . Starting with the two equations

$x = r \cos \theta$ and $y = r \sin \theta$ prove that the Normal acceleration of

P towards O is $r\omega^2$ where ω is the rate of change of θ with respect to time.

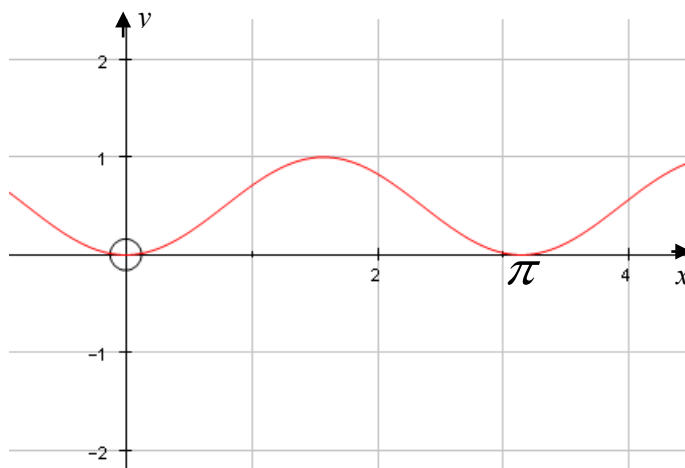
(a) Consider the polynomial $P(x) = x^4 - 4x^3 + 5x^2 - 2x - 2$.

- (i) Show that the curve $y = P(x)$ has a maximum turning point at $(1, -2)$ and minimum turning points at $x = 1 \pm \frac{1}{2}\sqrt{2}$. Hence deduce from a sketch of the curve that the equation $P(x) = 0$ has two real roots and two non-real roots. 3
- (ii) Explain why the real roots cannot be rational. What do you know about the nature of the non-real roots? 2
- (iii) Given that $1 + i$ is a root of the equation $P(x) = 0$, factor $P(x)$ into two quadratic factors with rational coefficients. Hence find the x -intercepts of the curve $y = P(x)$ and show them on your graph. 3

(b) (i) Prove that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$. 3

Hence show $\int_0^\pi x \cos 2x dx = 0$.

(ii)



The area bounded by the curve $y = \sin^2 x$ and the x -axis between $x = 0$ and $x = \pi$ is rotated through one revolution about the y -axis. By taking the limiting sum of the volumes of cylindrical shells, show that the volume of the solid of

revolution is given by $V = 2\pi \int_0^\pi x \sin^2 x dx$. Hence by using

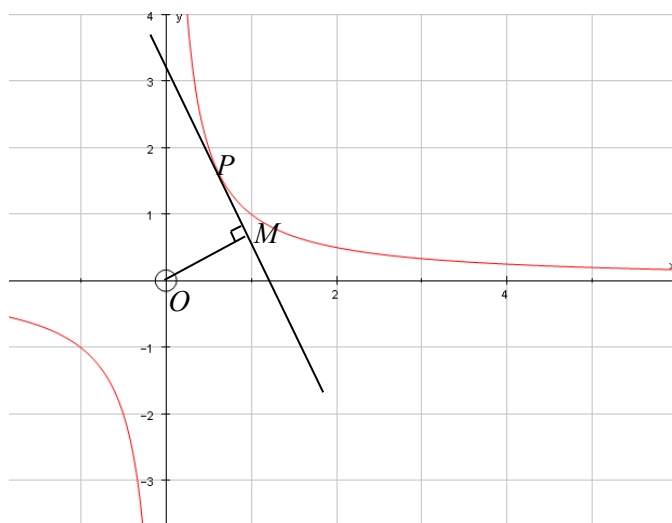
your answer in part (i) or otherwise find the volume of this solid.

P.T.O.....

Question 8 Use a SEPARATE writing booklet

Marks

- (a) A particle of mass M is projected vertically upward under gravity with speed U in a medium in which the resistance is Mk times the speed, here k is a positive constant. If the particle reaches its greatest height H in time T , show that $U = gT + kH$. (You may assume that the net force is given by $-Mg - Mkv$ with upward direction positive.) 4
- (b) $P(a \cos \theta, b \sin \theta)$ is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. PQ is a diameter of the ellipse. The tangent to the ellipse at P meets the vertical through Q at R and the Y -axis at V . O is the origin.
- (i) Prove that the tangent at P has equation $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$. 2
- (ii) Show that the area of $\triangle PQR = 4$ times the area of $\triangle POV$. 1
- (iii) Show that the area of $\triangle PQR$ is $\frac{2ab}{|\tan \theta|}$ square units. 2
- (c) $P(t, \frac{1}{t})$ is a variable point on the rectangular hyperbola $xy = 1$. M is the foot of the perpendicular from the origin to the tangent to the hyperbola at P .



- (i) Show that the tangent to the hyperbola at P has equation $x + t^2 y = 2t$. 2
- (ii) Find the equation of OM . 1
- (iii) Show that the locus of M as P varies has equation $x^2 + y^2 = 2\sqrt{xy}$. 3

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

HSC Trial Maths Ext 2 2008 Solutions Shore School 1

$$1(a) I = \int (e^{2x} + 2 + e^{-2x}) dx$$

$$= \frac{1}{2} e^{2x} + 2x - \frac{1}{2} e^{-2x} + c \quad (2)$$

$$(b) \text{ Let } \frac{2x^2 - 2x + 1}{(x-2)(x^2+1)} = \frac{a}{x-2} + \frac{bx+c}{x^2+1}$$

$$\text{For } x=2, \frac{5}{5} = a \Rightarrow a=1$$

$$\text{Also } 2x^2 - 2x + 1 = x^2 + 1 + (bx+c)(x-2)$$

$$= (1+b)x^2 - 2bx + cx + 1$$

$$\therefore b=1 \text{ and } c=0$$

$$\therefore I = \int \left(\frac{1}{x-2} + \frac{x}{x^2+1} \right) dx$$

$$= \ln|x-2| + \frac{1}{2} \ln|x^2+1| + c$$

$$\text{or } \ln|(x-2)\sqrt{x^2+1}| + c \quad (3)$$

$$(c) \text{ Let } t = \tan \frac{x}{2}$$

$$\therefore 2dt = \sec^2 \frac{x}{2} dx$$

$$= (1 + \tan^2 \frac{x}{2}) dx$$

$$= (1 + t^2) dx \quad \frac{dx}{2t} = \frac{dt}{1+t^2}$$

$$\therefore dx = \frac{2}{1+t^2} dt$$

$$\tan x = \frac{2t}{1-t^2}, \quad \sec x = \frac{2t}{1-t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$\therefore I = \int_0^1 \frac{1}{2 + \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{1}{1+t^2+2t} dt$$

$$= \int_0^1 \frac{1}{t^2+t+\frac{1}{4}+\frac{3}{4}} dt$$

$$= \int_0^1 \frac{1}{(t+\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} dt$$

$$= \left[\frac{2}{\sqrt{3}} \tan^{-1} \left((t+\frac{1}{2}) \cdot \frac{2}{\sqrt{3}} \right) \right]_0^1$$

$$= \frac{2}{\sqrt{3}} \left(\tan^{-1} \sqrt{3} - \tan^{-1} \frac{1}{\sqrt{3}} \right)$$

$$= \frac{\pi}{3\sqrt{3}} \quad (5)$$

$$1(d) I_n = \int_0^1 \frac{1}{(1+x^2)^n} \cdot d(x) \quad \left(\frac{1}{2} \text{ start} \right)$$

$$= \left[x \cdot \frac{1}{(1+x^2)^n} \right]_0^1 - \int_0^1 x d \left(\frac{1}{(1+x^2)^n} \right)$$

$$= \frac{1}{2^n} - \int_0^1 x \cdot \frac{-n \cdot 2x}{(1+x^2)^{n+1}} dx$$

$$= \frac{1}{2^n} + 2n \int_0^1 \frac{x^2}{(1+x^2)^{n+1}} dx$$

$$= \frac{1}{2^n} + 2n \int_0^1 \frac{(1+x^2) - 1}{(1+x^2)^{n+1}} dx$$

$$= \frac{1}{2^n} + 2n \cdot I_n - 2n \cdot I_{n+1}$$

$$\therefore 2n I_{n+1} = \frac{1}{2^n} + (2n-1) I_n$$

$$\therefore I_{n+1} = \frac{1}{n \cdot 2^{n+1}} + \frac{2n-1}{2n} I_n \quad (3)$$

$$I_3 = \frac{1}{2 \cdot 2^3} + \frac{3}{4} I_2$$

$$\& I_2 = \frac{1}{1 \cdot 2^2} + \frac{1}{2} I_1$$

$$\& I_1 = \int_0^1 \frac{1}{1+x^2} dx$$

$$= \left[\tan^{-1} x \right]_0^1$$

$$= \frac{\pi}{4}$$

$$\therefore I_3 = \frac{1}{16} + \frac{3}{4} \left(\frac{1}{4} + \frac{\pi}{8} \right)$$

$$= \frac{1}{4} + \frac{3\pi}{32}$$

$$= \frac{1}{32} (3\pi + 8) \quad (2)$$

$$2(a)(i) \frac{z_1}{z_2} = \frac{1+i}{\sqrt{3}-i} \times \frac{\sqrt{3}+i}{\sqrt{3}+i}$$

$$= \frac{\sqrt{3}-1 + (1+\sqrt{3})i}{3+1}$$

$$= \frac{\sqrt{3}-1}{4} + \frac{1+\sqrt{3}}{4}i \quad (1)$$

$$(ii) \begin{array}{l} \swarrow z_1 = \sqrt{2} \cos \frac{\pi}{4} \\ \searrow z_2 = 2 \cos \frac{-\pi}{6} \end{array} \quad (2)$$

$$(iii) \frac{z_1}{z_2} = \frac{\sqrt{2} \cos \frac{\pi}{4}}{2 \cos \frac{-\pi}{6}}$$

$$= \frac{1}{\sqrt{2}} \cos \frac{5\pi}{12}$$

Equate real parts in (i) & (iii)

$$\therefore \frac{1}{\sqrt{2}} \cos \frac{5\pi}{12} = \frac{\sqrt{3}-1}{4}$$

$$\therefore \cos \frac{5\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{1}{4}(\sqrt{6}-\sqrt{2}) \quad (1)$$

$$(iv) z_1(z_2)^n = \sqrt{2} \cos \frac{\pi}{4} \cdot 2^n \cos \frac{-n\pi}{6}$$

$$= \sqrt{2} \cdot 2^n \cos \left(\frac{\pi}{4} - \frac{n\pi}{6} \right)$$

For $z_1(z_2)^n$ to be real, arg is 11π ,
 11 integer.

$$\text{But } \frac{\pi}{4} - \frac{n\pi}{6} = \frac{3-2n}{12} \pi$$

$\therefore 3-2n$ has to be multiple of 12.

But $2n$ is always even $\therefore 3-2n$

is always odd

$\therefore$ it can never be a multiple of 12. (2)

$$(b)(i) \vec{OA} \text{ is } 2i(p+iz) = \underline{-2z + 2pi} \quad (1)$$

$$(ii) \vec{OB} \text{ is } -2z + 2pi + p + iz = \underline{(p-2z) + (2p+z)i} \quad (1)$$

$$(iii) \vec{OC} \text{ is } -(p-2z) + (2p+z)i + (p+iz) = \underline{+2z + 2pi}$$

$$\text{OR } \vec{BC} = -\vec{OA}$$

$$= \underline{2z - 2pi} \quad (1)$$

$$(iv) \vec{AC} \text{ is } (p+iz) - (-2z + 2pi) = \underline{p + 2z + i(z-2p)} \quad (1)$$

$$(i) (c) z^n = \cos n\theta + i \sin n\theta \quad z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$$

$$\therefore z^n + z^{-n} = \cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta)$$

$$= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta$$

$$= \underline{2 \cos n\theta} \quad (1)$$

$$z^n - z^{-n} = \cos n\theta + i \sin n\theta - \cos(-n\theta) + i \sin(-n\theta)$$

$$= \underline{2i \sin n\theta} \quad (1)$$

$$(ii) (z+z^{-1})^3 = z^3 + 3z + 3z^{-1} + z^{-3}$$

$$(z-z^{-1})^3 = z^3 - 3z + 3z^{-1} - z^{-3}$$

$$\therefore (z+z^{-1})^3 + (z-z^{-1})^3 = 2z^3 + 6z^{-1}$$

Let $z = \cos \theta + i \sin \theta$ and using (i) above

$$\therefore (2 \cos \theta)^3 + (2i \sin \theta)^3 = 2 \cos 3\theta + 6 \cos(-\theta)$$

$$\therefore 8 \cos^3 \theta - 8i \sin^3 \theta = 2 \cos 3\theta + 2i \sin 3\theta$$

$$+ 6 \cos \theta - 6i \sin \theta$$

Equate real parts:

$$8 \cos^3 \theta = 2 \cos 3\theta + 6 \cos \theta \quad (1)$$

Equate imaginary parts:

$$-8 \sin^3 \theta = 2 \sin 3\theta - 6 \sin \theta \quad (2)$$

(1) - (2):

$$8(\cos^3 \theta + \sin^3 \theta) = 2(\cos 3\theta - \sin 3\theta) + 6(\cos \theta + \sin \theta)$$

$$\therefore 4(\cos^3 \theta + \sin^3 \theta) = (\cos 3\theta - \sin 3\theta) + 3(\cos \theta + \sin \theta)$$

$$(3)$$

(b)(i) Assuming $|PA| = |OC|$

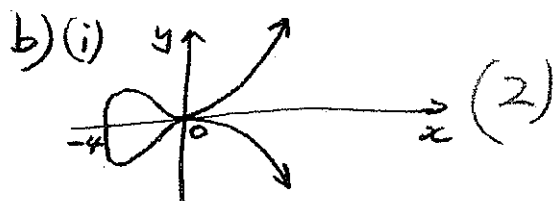
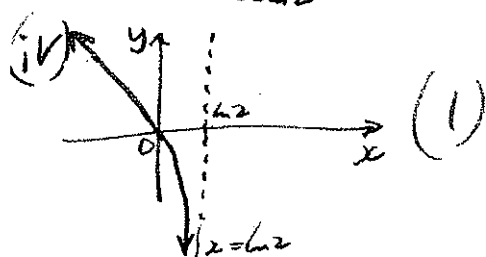
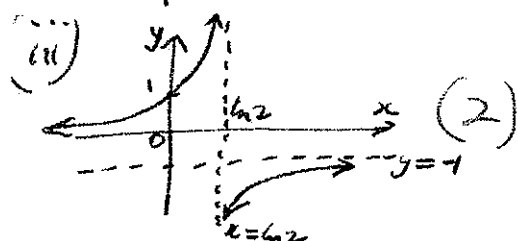
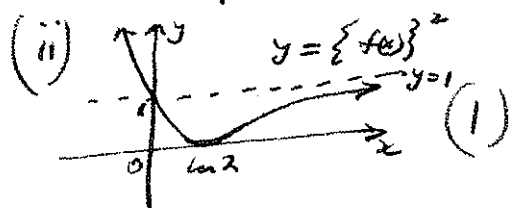
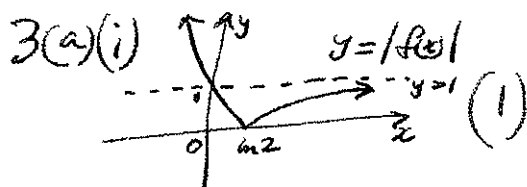
Then (i) $\vec{OA} = -z + ip$

$$(ii) \vec{OB} = (p-z) + i(p+z)$$

$$(iii) \vec{BC} = z - ip$$

$$(iv) \vec{AC} = (p+z) + i(z-p)$$

$$(3 \pm 4)$$



(ii) Area = $2 \int_{-4}^0 x \sqrt{4+x} dx$

Let $u = 4+x \therefore x = u-4$
 $du = dx \quad x^2 = u^2 - 8u + 16$

$\therefore \text{Area} = 2 \int_0^4 (u-4)^2 \sqrt{u} du$

$= 2 \int_0^4 \sqrt{u} (u^2 - 8u + 16) du$

$= 2 \int_0^4 u^{5/2} - 8u^{3/2} + 16u^{1/2} du$

$= 2 \left[\frac{2}{7} u^{7/2} - 8 \cdot \frac{2}{5} u^{5/2} + 16 \cdot \frac{2}{3} u^{3/2} \right]_0^4$

$= \frac{4096}{105} u^2 \quad (3) \text{ or } 39 \frac{1}{105}$

(c) (i) Let $y = x^2 \therefore x = \sqrt{y}$

$\therefore (\sqrt{y})^3 - 3(\sqrt{y})^2 - 2\sqrt{y} + 4 = 0$

$\therefore y\sqrt{y} - 3y - 2\sqrt{y} + 4 = 0$

$\therefore (y-2)\sqrt{y} = 3y - 4$

$\therefore (y-2)^2 \cdot y = (3y-4)^2$

$\therefore y^3 - 4y^2 - 9y^2 + 4y + 24y - 16 = 0$

$\therefore y^3 - 13y^2 + 28y - 16 = 0 \quad (2)$

(ii) $\therefore \alpha^2 + \beta^2 + \gamma^2 = -\frac{b}{a}$
 $= 13 \quad (1)$

(iii) Now $\alpha^3 - 3\alpha^2 - 2\alpha + 4 = 0$

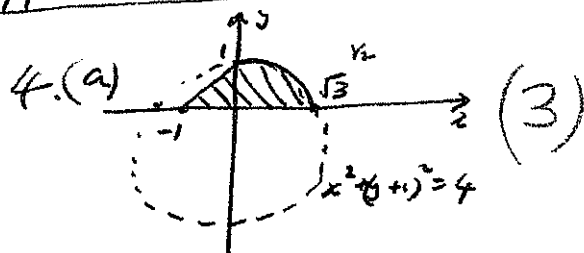
$\beta^3 - 3\beta^2 - 2\beta + 4 = 0$

$\gamma^3 - 3\gamma^2 - 2\gamma + 4 = 0$

$\therefore \alpha^3 + \beta^3 + \gamma^3 = 3(\alpha^2 + \beta^2 + \gamma^2) + 2(\alpha + \beta + \gamma) - 12$

$= 3(13) + 2(3) - 12$

$= 33 \quad (2)$



(b)(i) Let $z = r \cos \theta$, $z^5 = 1$

But $z^5 = r^5 \cos 5\theta$ by de Moivre's Th.

$\therefore r = 1$ & $\cos 5\theta = \cos 0, \cos 2\pi, \dots$

$\therefore 5\theta = 0, 2\pi, \dots$

$\therefore \theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \dots$

$\therefore z = 1, \cos \frac{2\pi}{5}, \cos \frac{4\pi}{5}, \cos \frac{6\pi}{5}, \cos \frac{8\pi}{5}$ (3)

(ii) $z^5 - 1 = (z-1)(z - \cos \frac{2\pi}{5}) \dots$

Now $z - \cos \frac{8\pi}{5} = (z - \cos(-\frac{2\pi}{5}))$
 $= (z - \cos \frac{2\pi}{5})$

Let $\alpha = \cos \frac{2\pi}{5}$ & consider

$(z - \alpha)(z - \bar{\alpha}) = z^2 - 2\operatorname{Re}(\alpha)z + \alpha\bar{\alpha}$

where $|\alpha| = 1 \therefore \alpha\bar{\alpha} = 1$

$\therefore (z - \cos \frac{2\pi}{5})(z - \cos \frac{8\pi}{5}) = z^2 - 2\cos \frac{2\pi}{5}z + 1$

$\therefore (z - \cos \frac{4\pi}{5})(z - \cos \frac{6\pi}{5}) = z^2 - 2\cos \frac{4\pi}{5}z + 1$ (3)

$\therefore z^5 - 1 = (z-1)(z^2 - 2\cos \frac{2\pi}{5}z + 1)(z^2 - 2\cos \frac{4\pi}{5}z + 1)$

(iii) $z^5 - 1 = 0$

$\therefore (z-1)(z^4 + z^3 + z^2 + z + 1) = 0$ (1)

$\therefore z = 1$ or $z = \cos \frac{2\pi}{5}, \cos \frac{4\pi}{5}, \cos \frac{6\pi}{5}, \cos \frac{8\pi}{5}$

(iv) Using sum of roots $= -\frac{1}{a}$

$\therefore \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} + \cos \frac{6\pi}{5} + \cos \frac{8\pi}{5} = -1$

Equating real parts:

$\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} + \cos \frac{6\pi}{5} + \cos \frac{8\pi}{5} = -1$

But $\cos \theta = \cos(-\theta)$

$\therefore 2\cos \frac{2\pi}{5} + 2\cos \frac{4\pi}{5} = -1$

$\therefore \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$ (2)

(c) RTP $\frac{1}{x} + \frac{1}{y} - \frac{4}{x+y} \geq 0$

Now $\frac{1}{x} + \frac{1}{y} - \frac{4}{x+y} = \frac{z+y}{xy} - \frac{4}{x+y}$
 $= \frac{x^2 + 2xy + y^2 - 4xy}{xy(x+y)}$

$= \frac{x^2 - 2xy + y^2}{xy(x+y)}$

$= \frac{(x-y)^2}{xy(x+y)}$

Now $x, y > 0 \therefore xy > 0, x+y > 0$ & $(x-y)^2 > 0$

$\therefore \frac{1}{x} + \frac{1}{y} - \frac{4}{x+y} > 0$

$\therefore \frac{1}{x} + \frac{1}{y} \geq \frac{4}{x+y}$ (3)

Alternative proof:

Consider $(a-b)^2 > 0$

$\therefore a^2 + b^2 \geq 2ab$

$\therefore u + v \geq 2\sqrt{uv}$ where $u = a^2$
 $v = b^2$

$\therefore \frac{1}{x} + \frac{1}{y} \geq \frac{2}{\sqrt{xy}}$ (A)

But $x+y \geq 2\sqrt{xy}$

$\therefore \frac{1}{x+y} \leq \frac{1}{2\sqrt{xy}}$

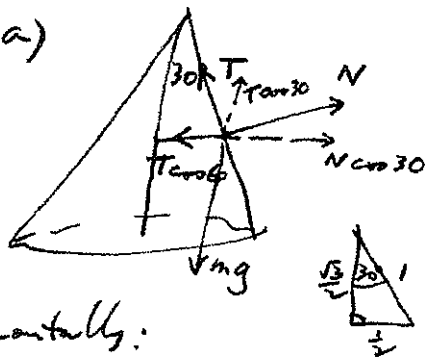
$\therefore \frac{2}{x+y} \leq \frac{1}{\sqrt{xy}}$

From (A) $\frac{1}{x} + \frac{1}{y} \geq 2 \times \frac{2}{x+y}$

$\therefore \frac{1}{x} + \frac{1}{y} \geq \frac{4}{x+y}$

(b)(i) Ans only + circle drawn. $1\frac{1}{2}/3$

5. (a)



(i)

Horizontally:

$$F = T \cos 60^\circ - N \cos 30^\circ$$

$$\therefore \frac{m v^2}{r} = \frac{1}{2} T - \frac{\sqrt{3}}{2} N \quad (3)$$

$$\text{But } v = 1 \text{ and } r = \frac{1}{2}$$

$$\therefore 6 = \frac{1}{2} T - \frac{\sqrt{3}}{2} N$$

$$\therefore 12 = T - \sqrt{3} N \quad (1)$$

Vertically:

$$F = T \sin 30^\circ + N \sin 60^\circ - mg$$

$$\therefore 0 = \frac{\sqrt{3}}{2} T + \frac{1}{2} N - 3 \times 10$$

$$\therefore \sqrt{3} T + N = 60$$

$$\therefore 3T + \sqrt{3} N = 60\sqrt{3} \quad (2)$$

Add: (1) + (2)

$$4T = 12 + 60\sqrt{3}$$

$$\therefore T = 3 + 15\sqrt{3} \approx 28.98$$

Subs back into (1):

$$12 = 3 + 15\sqrt{3} - \sqrt{3} N$$

$$\therefore N = \frac{15\sqrt{3} - 9}{\sqrt{3}}$$

$$\therefore N = 15 - 3\sqrt{3} \approx 9.8038 \quad (4)$$

(ii) Let $N = 0$ for critical value

$$\text{Subs into (3)} \therefore \frac{3v^2}{\frac{1}{2}} = \frac{1}{2} T$$

$$\therefore T = 12v^2$$

$$\text{Subs into (2)} \therefore T = 20\sqrt{3}$$

$$\therefore 20\sqrt{3} = 12v^2$$

$$\therefore v^2 = \frac{5\sqrt{3}}{3}$$

$$\therefore v = \sqrt{\frac{5\sqrt{3}}{3}} \approx \left(\frac{25}{3}\right)^{1/4} \approx 1.699 \quad (4)$$

(b) $t=0, x=0, v=0$
(i) $\uparrow kx \downarrow mg$
positive direction downwards.

$$F = mg - kx$$

$$\therefore m v \cdot \frac{dv}{dx} = mg - kx$$

$$\text{Now } m = 1$$

$$\therefore v \frac{dv}{dx} = g - kx$$

$$\therefore \int_0^x v dv = \int_0^x (g - kx) dx$$

$$\therefore \left[\frac{v^2}{2} \right]_0^x = \left[gx - \frac{1}{2} kx^2 \right]_0^x$$

$$\therefore \frac{v^2}{2} = gx - \frac{1}{2} kx^2$$

$$\therefore v^2 = x(2g - kx)$$

$$\therefore v = 0 \Rightarrow x = 0 \text{ or}$$

$$x = \frac{2g}{k} \quad (2)$$

(ii) from (i) $v^2 = x(2g - kx)$

$$\therefore \frac{dx}{dt} = \sqrt{2gx - kx^2}, v > 0$$

$$\therefore \int dt = \int \frac{1}{\sqrt{2gx - kx^2}} dx$$

$$\therefore \int_0^T \sqrt{k} dt = \int_0^x \frac{1}{\sqrt{-(x^2 - \frac{2gx}{k} + \frac{g^2}{k^2} - \frac{g^2}{k^2})}} dx$$

$$= \int_0^x \frac{dx}{\sqrt{\left(\frac{x}{k}\right)^2 - \left(x - \frac{g}{k}\right)^2}}$$

$$\therefore \left[\sqrt{k} t \right]_0^T = \left[\sin^{-1} \left(\frac{x - \frac{g}{k}}{\frac{g}{k}} \right) \right]_0^x$$

$$\therefore \sqrt{k} T = \sin^{-1} \left(\frac{k}{g} \left(x - \frac{g}{k} \right) \right) - \sin^{-1}(-1)$$

$$\therefore \sqrt{k} T = \sin^{-1} \left(\frac{k}{g} \left(x - \frac{g}{k} \right) \right) + \frac{\pi}{2}$$

$$\therefore \sin \left(\sqrt{k} T - \frac{\pi}{2} \right) = \frac{k}{g} \left(x - \frac{g}{k} \right)$$

$$\therefore x = \frac{g}{k} + \frac{g}{k} \sin \left(\sqrt{k} T - \frac{\pi}{2} \right)$$

$$= \frac{g}{k} \left(\sin \left(\frac{\sqrt{k} T - \pi}{2} \right) + 1 \right) \quad (5)$$

HSC Trial Maths Ext 2

6(a)(i)

$$\begin{aligned} LHS &= (1 + \cos \theta) + i \sin \theta \\ &= 2 \cos^2 \frac{\theta}{2} + i 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \\ &= 2 \cos \frac{\theta}{2} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2}) \\ &= RHS \quad (2) \end{aligned}$$

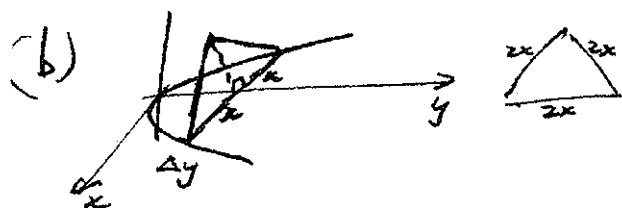
$$\begin{aligned} (ii) \quad 1 + z^2 &= 1 + \cos 2\theta \\ &= 1 + \cos 2\theta + i \sin 2\theta \\ &= 2 \cos \theta (\cos \theta + i \sin \theta) \text{ from (i)} \quad (1) \end{aligned}$$

$$\therefore 1 + z^2 = 2 \cos \theta \cdot z$$

$\therefore O, A, B$ lie on the same straight line with $2 \cos \theta$ as the scalar quantity. For B to lie on OA , $0 < 2 \cos \theta < 1$

$$\therefore 0 < \cos \theta < \frac{1}{2}$$

$$\therefore \frac{\pi}{3} < \theta < \frac{\pi}{2} \text{ \& } -\frac{\pi}{2} < \theta < -\frac{\pi}{3} \quad (2)$$



$$\begin{aligned} \Delta V &= \left(\frac{1}{2} (2x) (2x) \sin \frac{\pi}{3} \right) \Delta y \\ &= \sqrt{3} x^2 \Delta y \end{aligned}$$

$$\therefore V = \lim_{\Delta y \rightarrow 0} \sqrt{3} x^2 \Delta y$$

$$= \int_0^{16} \sqrt{3} x^2 dy$$

$$= \int_0^{16} \sqrt{3} y dy$$

$$= \left[\frac{\sqrt{3} y^2}{2} \right]_0^{16}$$

$$= 128 \sqrt{3} \quad (5)$$

$$(c) \quad x = r \cos \theta$$

$$\therefore \dot{x} = \frac{d}{dt} r \cos \theta \cdot \frac{d\theta}{dt}$$

$$= -r \sin \theta \cdot \omega$$

$$\therefore \ddot{x} = \omega \left(\frac{d}{dt} r \sin \theta \cdot \frac{d\theta}{dt} \right) + r \sin \theta \cdot \dot{\omega}$$

$$= \omega \cdot r \cos \theta \cdot \omega - r \sin \theta \cdot \dot{\omega}$$

$$= -r (\cos \theta \cdot \omega^2 + \sin \theta \cdot \dot{\omega})$$

2008 Solutions

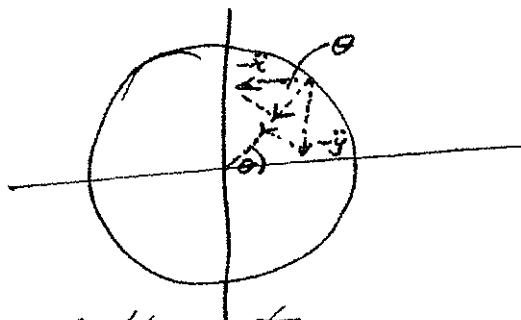
6

$$\text{Also } y = r \sin \theta$$

$$\therefore \frac{dy}{dt} = \frac{d}{dt} r \sin \theta \cdot \frac{d\theta}{dt}$$

$$\therefore \dot{y} = r \cos \theta \cdot \omega$$

$$\begin{aligned} \therefore \ddot{y} &= \omega \left(\frac{d}{dt} r \cos \theta \cdot \frac{d\theta}{dt} \right) + r \cos \theta \cdot \dot{\omega} \\ &= -r \sin \theta \cdot \omega^2 + r \cos \theta \cdot \dot{\omega} \end{aligned}$$



towards the centre

$$N = -\ddot{x} \cos \theta - \ddot{y} \sin \theta$$

$$= r (\cos^2 \theta \omega^2 + \sin^2 \theta \dot{\omega}^2 - r \cos \theta \cdot \sin \theta \cdot \dot{\omega})$$

$$= r \omega^2 (\sin^2 \theta + \cos^2 \theta)$$

$$= r \omega^2 \text{ as } \sin^2 \theta + \cos^2 \theta \equiv 1.$$

(5)

$$7(a)(i) P'(x) = 4x^3 - 12x^2 + 10x - 2$$

$$\begin{array}{r} 4x^3 - 12x^2 + 10x - 2 \\ \underline{4x^3 - 8x^2 + 2} \\ 4x^2 - 4x^2 + 10x - 2 \\ \underline{4x^2 - 8x + 2} \\ 2x - 2 \\ \underline{2x - 2} \\ 0 \end{array}$$

$$\text{Let } 4x^2 - 8x + 2 = 0$$

$$\therefore 2x^2 - 4x + 1 = 0$$

$$\therefore x = \frac{4 \pm \sqrt{16-8}}{4}$$

$$= \frac{2 \pm \sqrt{2}}{2}$$

$$= 1 \pm \frac{1}{2}\sqrt{2}$$

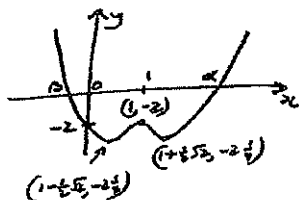
$$\therefore \text{stationary at } 1, 1 \pm \frac{1}{2}\sqrt{2}$$

$$P''(x) = 12x^2 - 24x + 10$$

$$\text{Now } P''(1) = -2 < 0 \therefore x=1 \text{ must be p.}$$

$$\& P''(1 \pm \frac{1}{2}\sqrt{2}) = 4 > 0 \therefore \text{min \& p.}$$

$$\& P''(-\frac{1}{2}\sqrt{2}) = 4 > 0 \therefore \text{min \& p.}$$



from the sketch d, b shown
are 2 real values.

As only 2, 2 must be unreal.

as $P(x)$ is of degree 4 (3)

(ii) $P(x)$ has integer coeffs, coeff
of x^4 is 1 & constant term is -2

$\therefore$ product of rational roots is -2

$\therefore$ test $x = 0, 1, -1, 2, -2$ into $P(x)$

$$P(0) = -2, P(1) = 10, P(-1) = -2, P(2) = 2, P(-2) = 70$$

$\therefore$ none are zero

$\therefore$ no rational roots (1)

(ii) (Cont).

Complex roots are conjugates of
each other as all the coeffs are real

(iii) roots are $1+i, 1-i, \alpha, \beta$

$$\text{Now } (x-(1+i))(x-(1-i)) = (x-1-i)(x-1+i)$$

$$= (x-1)^2 - i^2$$

$$= x^2 - 2x + 1 + 1$$

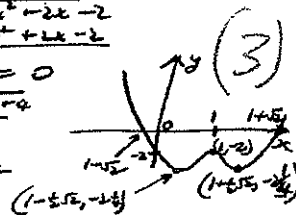
$$= x^2 - 2x + 2$$

$$\begin{array}{r} x^2 - 2x + 2 \quad x^3 - 4x^2 + 5x^2 - 2x - 2 \\ \underline{x^3 - 2x^2 + 2x} \\ -2x^2 + 3x^2 - 2x \\ \underline{-2x^2 + 4x^2 - 4x} \\ -x^2 - 2x - 2 \\ \underline{-x^2 + 2x + 2} \\ 0 \end{array}$$

$$\text{Solving } x^2 - 2x - 1 = 0$$

$$\therefore x = \frac{2 \pm \sqrt{4+4}}{2}$$

$$= 1 \pm \sqrt{2}$$



$$(b)(i) LHS = [F(x)]_0^{\pi} \text{ since } F(x) = f(x)$$

$$= F(\pi) - F(0)$$

$$RHS = [-F(\pi-x)]_0^{\pi}$$

$$= -F(\pi) + F(0)$$

$$\therefore LHS = RHS \quad (1)$$

$$\text{Let } a = \pi, f(x) = x \cos 2x, f(\pi-x) = (\pi-x) \cos 2(\pi-x)$$

$$\therefore \int_0^{\pi} x \cos 2x dx = \int_0^{\pi} (\pi-x) \cos 2(\pi-x) dx$$

$$= \int_0^{\pi} \pi \cos(2\pi-2x) dx - \int_0^{\pi} x \cos(2\pi-2x) dx$$

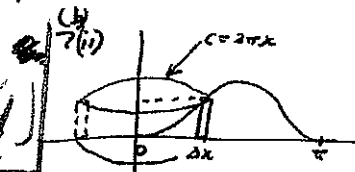
$$= \int_0^{\pi} \pi \cos 2x dx - \int_0^{\pi} x \cos 2x dx$$

$$\therefore 2 \int_0^{\pi} x \cos 2x dx = \pi \int_0^{\pi} \cos 2x dx$$

$$= \pi \left[\frac{\sin 2x}{2} \right]_0^{\pi}$$

$$= 0$$

$$\therefore \int_0^{\pi} x \cos 2x dx = 0 \quad (2)$$



Opening out the cylindrical shell



$$\therefore \Delta V = 2\pi r y \Delta x$$

$$= 2\pi r x \cdot \frac{1}{2} \Delta x$$

$$\therefore V = \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n 2\pi r x_i \sin^2 x_i \Delta x$$

$$= \int_0^{\pi} 2\pi r x \sin^2 x dx$$

$$= 2\pi \int_0^{\pi} x \left(\frac{1 - \cos 2x}{2} \right) dx$$

$$= 2\pi \int_0^{\pi} \frac{x}{2} dx - 2\pi \int_0^{\pi} \frac{x \cos 2x}{2} dx$$

$$= \pi \left[\frac{x^2}{2} \right]_0^{\pi} - 0 \text{ by (i) above}$$

$$= \frac{\pi^3}{2} \quad (4)$$

HSC Trial Maths Ext 2

$$x=4, v=9, t=T \quad \downarrow M_y \quad \downarrow M_{kv}$$

$$f(a) \quad \uparrow \quad T=0, v=4, x=0$$

$$F = -M_y - M_{kv}$$

$$\therefore M \cdot \frac{dv}{dx} = -M_y - M_{kv}$$

$$\therefore \int \frac{v dv}{g+kv} = \int -dx$$

$$\therefore \int \frac{g+kv-g}{g+kv} dv = \int -k dx$$

$$\therefore \int \left(1 - \frac{g}{k} \frac{k}{g+kv}\right) dv = \int -k dx$$

$$\therefore \left[v - \frac{g}{k} \ln(g+kv)\right]_u^4 = -kH$$

$$\therefore -\frac{g}{k} \ln g - u + \frac{g}{k} \ln(g+ku) = -kH \quad (1)$$

$$\text{Also } M \cdot \frac{dv}{dt} = -M_y - M_{kv}$$

$$\therefore \int \frac{dv}{g+kv} = \int -dt$$

$$\therefore \left[\ln(g+kv)\right]_u^0 = \left[-kt\right]_0^T$$

$$\therefore -\ln(g+ku) + \ln g = -kT$$

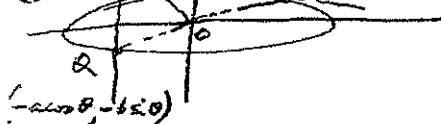
$$\therefore \ln(g+ku) = kT + \ln g \quad (2)$$

Sub into (1)

$$\therefore -\frac{g}{k} \ln g - u + \frac{g}{k} (kT + \ln g) = -kH$$

$$\therefore u = gT + kH \quad (4)$$

$$(b) \quad R(-a \cos \theta, b \sin \theta) \quad V(a \cos \theta, b \sin \theta)$$



$$(i) \quad x = a \cos \theta, \quad y = b \sin \theta$$

$$\therefore \frac{dx}{d\theta} = -a \sin \theta, \quad \frac{dy}{d\theta} = b \cos \theta$$

$$\therefore \frac{dy}{dx} = \frac{b \cos \theta}{-a \sin \theta}$$

$$\therefore \text{tangent } \frac{y - b \sin \theta}{x - a \cos \theta} = \frac{-b \cos \theta}{a \sin \theta}$$

$$\therefore \frac{y \sin \theta - a b \sin^2 \theta}{x \sin \theta - a b \cos^2 \theta} = \frac{-b \cos \theta}{a \sin \theta}$$

$$\therefore b x \cos \theta + a y \sin \theta = a b (\sin^2 \theta + \cos^2 \theta)$$

$$\therefore \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 \quad (2)$$

2008 Solution:

(8)

$$(b)(ii) \text{ area } \Delta PQR = \frac{1}{2} \cdot PQ \cdot PR \cdot \sin \angle QPR$$

$$\& \text{ area } \Delta POV = \frac{1}{2} \cdot PO \cdot PV \cdot \sin \angle OPV$$

$$\therefore \text{RTP } PO \cdot PR = 4 PO \cdot PV$$

Now $PQ = 2 \cdot PO$ (diameter = twice radii)

& $PR = 2 PV$ (ratio of intercepts on parallel lines)

$$\therefore PQ \cdot PR = 4 PO \cdot PV \text{ and area } \Delta PQR = 4 \times \text{area } \Delta POV \quad (1)$$

$$(iii) \text{ To find } V: \frac{x \cos \theta + y \sin \theta}{a} = 1 \text{ & } x = 0$$

$$\therefore y = \frac{b}{\sin \theta} \therefore V(0, \frac{b}{\sin \theta})$$

$$\therefore \text{area } \Delta POV = \left| \frac{1}{2} \times \frac{b}{\sin \theta} \times a \cos \theta \right| \text{ using } \theta = \frac{1}{2} \angle H$$

$$= \frac{ab}{2(1+\cos \theta)}$$

$$\therefore \text{area of } \Delta PQR = \frac{2ab}{1+\cos \theta} \text{ using (i) above.} \quad (2)$$

$$(c)(i) \quad x = t \text{ and } y = t^{-1}$$

$$\therefore \frac{dx}{dt} = 1 \text{ & } \frac{dy}{dt} = \frac{-1}{t^2}$$

$$\therefore \frac{dy}{dx} = -\frac{1}{t^2}$$

$$\therefore \text{tangent } \frac{y - \frac{1}{t}}{x - t} = -\frac{1}{t^2}$$

$$\therefore t^2 y - t = -x + t$$

$$\therefore x + t^2 y = 2t \quad (2)$$

$$(ii) \text{ grad of tangent} = \frac{1}{t^2}$$

$$\therefore \text{grad of OM} = t^2$$

$$\text{eqn } \frac{y-0}{x-0} = t^2$$

$$\therefore y = t^2 x \quad (1)$$

$$(iii) \text{ to find M: Solve } y = xt^2$$

$$y = \frac{1}{t^2} x + \frac{2}{t}$$

$$\therefore xt^2 = -\frac{1}{t^2} x + \frac{2}{t}$$

$$\therefore xt^4 = -x + 2t$$

$$\therefore x = \frac{2t}{t^4+1}$$

$$\text{Sub back, } y = \frac{2t}{t^4+1} \times t^2$$

$$\therefore M\left(\frac{2t}{t^4+1}, \frac{2t^3}{t^4+1}\right)$$

$$\text{RTS } x^2 + y^2 = 2 \sqrt{xy}$$

$$\text{Sub } LHS = x^2 + y^2$$

$$= \frac{4t^2}{t^4+1} + \frac{4t^6}{t^4+1} = \frac{4t^2}{t^4+1} \quad (3)$$

$$\begin{aligned} RHS &= 2\sqrt{xy} \\ &= 2\sqrt{\frac{2t}{t^4+1} \times \frac{2t^3}{t^4+1}} \\ &= \frac{4t^2}{t^4+1} \\ &= LHS \end{aligned}$$